Velocity and position graphs

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Learning target: Connect a position-time graph to velocity using slope.

Suggested first pass: 1, 2, 3, 9. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 2; 1 course days later.*
Distance 12 m; displacement +2 m (right); final position 5 m.

02. Explain a worked example (worked)
Straight line through (0, 1) and (3, 7), rise 6 m and run 3 s; positive slope.

03. Read the motion picture (model)
x(0)=1 m and x(3)=7 m; motion is right/positive.

04. Find the rate (practice)
7-1=6 m; 3 s; +2 m/s/right.

05. A pause (reason)
No. Position stays 5 m, so displacement is zero and velocity is zero.

06. Draw a walk with a pause (reason)
0-2 s: right at 2 m/s; 2-4 s: stopped; 4-6 s: left at 2 m/s. Points (0, 1),(2, 5),(4, 5),(6, 1).

07. Distance or displacement? (reason)
Distance 4+4=8 m; displacement 0 m. Average velocity uses displacement; the walker returns to the start.

08. Choose a method (mixed) *Review: Unit 1, Day 2; 1 course days later.*
Distance 14 m; displacement +2 m (right); final position 6 m.

09. Exit check (exit)
6 m left in 3 s = 2 m/s left, or -2 m/s with right positive. Straight descending line from (0, 9) to (3, 3).

10. STRETCH: Apply it another way (stretch)
Any two parallel lines with slope 2 m/s; vertical intercepts differ.

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01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 2; 1 course days later.*
Distance 13 m; displacement -3 m; final position +2 m.

02. Explain a worked example (worked)
Straight line through (0, 1) and (3, 7), rise 6 m and run 3 s; positive slope.

03. Interpret all three intervals (model)
0-2 s: (6-2)/2=+2 m/s; 2-4 s: 0; 4-6 s: (0-6)/2=-3 m/s.

04. Compare two averages (practice)
Distance 4+6=10 m; average speed 10/6=1.67 m/s. Displacement 0-2=-2 m; average velocity -0.333 m/s.

05. A line that misses the origin (reason)
v=-2 m/s; x=18-2 t (m, s); x(9)=0 m, assuming the same line continues.

06. Construct the graph (reason)
Vertices (0,-2),(2, 4),(4, 4),(7,-2). Net displacement zero; slope signs match each interval.

07. Separate height and slope (reason)
It gives average velocity over 0-5 s if x(0)=0 m. It is also instantaneous velocity at 5 s only if the appropriate local slope is 2 m/s, e.g. a straight line from the origin. A single point is insufficient.

08. Choose a method (mixed) *Review: Unit 1, Day 2; 1 course days later.*
Distance 15 m; displacement +1 m; final position -2 m.

09. Exit check (exit)
v=8/4=+2 m/s; position increases by 2 m per second. x(7)=10 m.

10. STRETCH: Apply it another way (stretch)
1+2 t=9-t gives t=8/3 s and x=19/3 m. Intersection represents same place and time.